

Diagnosing Autism Spectrum Disorder Based on the Brain's Functional Connectome

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Abstract: Autism Spectrum Disorder (ASD) diagnosis currently depends on subjective behavioral assessments, motivating the search for objective neuroimaging biomarkers. This study evaluates different types of connectivity measures from Resting-State Function Magnetic Resonance Imaging (rs-fMRI) to diagnose ASD in a large multisite dataset. Different Functional Connectivity (FC) measures were computed between the time series corresponding to resting state networks. FC was obtained via the Pearson and Spearman Correlation Coefficient (PCC and SCC), but also through a non-linear lag-independent method called Dynamic Time Warping (DTW). Feature vectors were input to five different classifiers, obtaining their performance matrices, which were assessed using nested cross-validation to prevent data leakage. The highest classification accuracy (65%) was achieved using DTW features on a homogeneous sub dataset, indicating that non-linear correlation may outperform linear methods under consistent scan protocols. Compared to prior studies that often rely solely on linear correlation methods and use relatively small datasets, this research highlights the importance of robust validation, non-linear metrics and consistent scanning protocols in developing reliable ASD classifiers from rs-fMRI.

1 INTRODUCTION

Autism Spectrum Disorder (ASD) is a neuro-developmental condition characterized by deficits in social interaction, communication, and restrictive, repetitive patterns of behavior. Early diagnosis of this condition is essential to enable timely intervention and effective treatment. Currently, ASD identification often involves behavioral observation, which in some cases can be prolonged and may require the expertise of a multidisciplinary team of highly trained specialists. However, in some cases, limited understanding of the underlying neuropathology may lead to suboptimal treatment outcomes, as the current diagnosis remains subjective while an objective one is desired [Sólon et al., 2017] [Insel et al., 2010].

Functional Magnetic Resonance Imaging (fMRI), an imaging technique that measures brain activity,

has gained significant popularity in recent decades for investigating potential differences between individuals with ASD and Healthy Controls (HC) [Santana et al., 2022] [Schielen et al., 2024b]. Its ability to monitor brain activity with sufficiently high spatial resolution in a safe and noninvasive manner plays a key role driving its widespread adoption [Poldrack et al., 2011]. A variety of studies seek to identify a neural biomarker for ASD by measuring the byproduct of neural activity, the concentration of deoxygenated hemoglobin known as the Blood-Oxygen-Level-Dependent (BOLD) signal.

Several approaches aim to identify such a biomarker, commonly through Functional Connectivity (FC). FC is a connectivity method which describes the statistical dependency between the activity of two brain regions.

FC can be assessed using various methods, with Pearson's Correlation Coefficient (PCC) being the most commonly used method in diagnostic MRI research [Santana et al., 2022] [Schielen et al., 2024b]. The PCC method assumes synchronization and measures the linear relationship between two time series but does not account for time lags or variations in

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speed between them. In addition, PCC assumes wide-sense stationarity, meaning that the mean and the covariance between the input signals remain unchanged over the course of the experiment. The stationarity of fMRI signals can be challenged, especially in the common event of head motion [Laumann et al., 2017]. Another method examined is Spearman’s Correlation Coefficient (SCC), a rank-based correlation method that does not assume stationarity or linearity, allowing it to identify linear and non-linear correlations between time series whose statistical properties can vary with time. SCC captures monotonic trends, but it does not necessarily account for temporal misalignments.

In contrast, Dynamic Time Warping (DTW) is a non-linear correlation method, which estimates the cost of warping one signal to align with another and therefore accommodates temporal misalignments, shifts, and non-stationary signals. The latter deserves closer attention as the assumption of stationarity is often not satisfied in fMRI [Muhe-Aldin et al., 2014]. This motivates the relatively novel computations of DC through DTW.

This research aims to compare the use of the PCC, SCC, and DTW methods to evaluate which method more efficiently improves ASD classification, using the Autism Brain Imaging Data Exchange (ABIDE) dataset [Martino, 2017] [Martino, 2014] [Schielen et al., 2024a].

This paper makes three key contributions: first, it evaluates the use of DTW as a non-linear method for FC, challenging the commonly used PCC for FC. It is also compared to SCC, which is similar to PCC but does not assume linearity. Second, it examines the impact of dataset heterogeneity on classification performance by comparing results across homogeneous and heterogeneous data subsets, and demonstrates that DTW outperforms other correlation methods in a large homogeneous subset. Lastly, it offers a potential framework for the clinical interpretation of the classification results.

The remainder of the paper starts with a state-of-the-art review in section 2, followed by the methodology applied in this research in Section 3. The results are discussed in Section 4, with the drawn conclusions provided in Section 5.

2 RELATED WORK

In this section, the current literature approaches to classifying ASD with fMRI are discussed. FC is expressed between the time series of two regions of interest (ROIs). Two ways of obtaining these exist: a

predefined atlas and a parcellation directly obtained on the studied sample with group ICA. The use of Independent Component Analysis (ICA) to identify ROIs and remove low-frequency noise and motion effects at Resting-State fMRI (rs-fMRI) has been successfully demonstrated and applied by different researchers [Beckmann et al., 2005, Beckmann et al., 2009, Schielen et al., 2024a]. To express statistical relationships between ROIs, various studies have relied almost entirely on PCC, assuming only synchronized BOLD signals indicate FC between two regions. Over the past decade, various studies have attempted to classify based on this FC using traditional machine learning classifiers [Yang et al., 2019, Yang et al., 2020, Biswal et al., 1995]. The focus is on the methods of interest to this paper - PCC, SCC, and DTW.

A 2019 study reported that a Ridge classifier achieved an accuracy of 71.98% using a 5-fold cross-validation and the Craddock 400 (CC400) brain atlas [Yang et al., 2019]. Cao et al. (2021) [Cao et al., 2021] have used PCC to extract FC with a graph convolutional network classifier and a 10-fold cross validation, resulting in accuracy of 73.3%. However, the listed studies applied hyperparameter tuning including the test set, which may cause data leakage and lead to overoptimistic results [Rosenblatt et al., 2024]. A research from 2020 employing a deep neural network for classification attained a higher accuracy of 75.27% with a 5-fold cross-validation [Yang et al., 2020]. More recently, Li et al. [Li et al., 2022] (2022) suggested that a transformer-based model achieved a classification accuracy of 74.18% with 10-fold cross-validation.

The aforementioned studies used PCC to assess FC with the ABIDE dataset, including more than 1000 scans of ASD and HC [Santana et al., 2022] [Biswal et al., 1995]. PCC accounts only for the linear correlation of different time series, which may prevent reliable identification of time series that share a non-linear correlation pattern. Therefore, other correlation methods may outperform PCC. For instance, a study from 2022 suggests that SCC (a non-linear correlation method) may be more applicable than PCC [Yang et al., 2022]. Furthermore, brain atlases typically contain many ROIs which scales quadratically with the number of features in pairwise connectivity. This might convolute direct clinical interpretation.

A different method is DTW, a powerful non-linear approach that compares the similarity of time series with varying speeds. Linke et al. [Linke et al., 2020] state that using the non-linear DTW method has detected more FC differences and improved the estimation of resting-state FC in ASD. However, the study utilized a relatively small sample size (fewer than 200

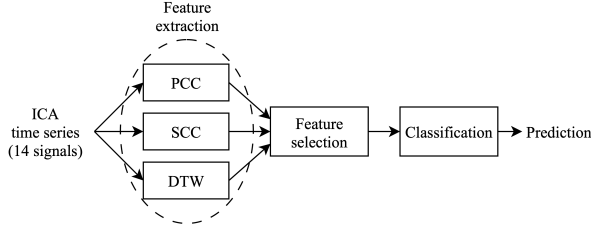


Figure 1: The suggested block scheme of the proposed approach in which ICA time series are input and extracts three types of features. These are analyzed in feature selection and classification. (ICA - Independent Component Analysis; PCC - Pearson’s Correlation Coefficient; SCC - Spearman’s Correlation Coefficient; DTW - Dynamic Time Warping)

participants) divided into three distinct age groups. Consequently, generalizability remains to be tested on a larger sample.

In summary, many recent studies have delivered promising results using FC. These studies attempted to diagnose ASD with different methodological approaches to feature extraction techniques and classification. While the effects of the feature extraction techniques and the choice of classifier vary across studies, all the aforementioned researchers agree on testing different algorithms and evaluating their performance on accuracy, sensitivity, and specificity through k-fold cross-validation [Rosenblatt et al., 2024]. Important to note is that FC was based on PCC in all these studies, which involves adopting its underlying assumptions. It remains unclear whether considering only linear correlation misses important information about coactivation patterns between time series.

3 METHODOLOGY

To predict ASD, the suggested approach is shown in Fig. 1. The system takes as an input the ICA time series (14 signals) and provides a binary prediction as an output.

3.1 Data Set

A preprocessed version of the ABIDE dataset was used in this study and comprises 900 participants: 417 subjects are diagnosed with ASD (361 males and 56 females; mean age 22 years) and 483 subjects are diagnosed as HC (377 males and 106 females; mean age 27 years) [Schielen et al., 2024a]. A subset of the data, consisting of 260 participants originating entirely from NYU Langone Health system, is examined. The subset represents the largest dataset pro-

viding a homogeneous scanning protocol, potentially leading to improved classification performance and reliability.

Initially, the rs-fMRI data were preprocessed to ensure slice-timing alignment, spatial normalization, smoothing and noise reduction. Then, a group ICA was performed on the fMRI data, where the components corresponding to the RSNs were selected with the Smith atlas as an initial reference [Beckmann et al., 2009]. This procedure yielded spatial maps, which then served as a design matrix in a multiple linear regression to derive subject-specific time series [Nickerson et al., 2017]. Each time series is constructed from 146 samples, as illustrated in Fig. 2.

The study’s results are focused on 14 of the 32 identified independent components, examining their inter-network correlation patterns to improve ASD classification and to explore their potential clinical significance. The 14 components and their labeling are shown in Fig. 3.

3.2 Feature Extraction

Both linear and non-linear correlation methods were considered to characterize brain connectivity patterns. The motivation behind the choice of extracting the PCC, SCC, and DTW features stems from considering the implicit assumption of linearity in PCC and investigating if a non-linear method can outperform. The initial part of the system is extracting the time series’ correlation features using PCC, SCC, and DTW. The calculation of r_{xy} , the PCC between 2 time series, is obtained as follows:

$$r_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}}, \quad (1)$$

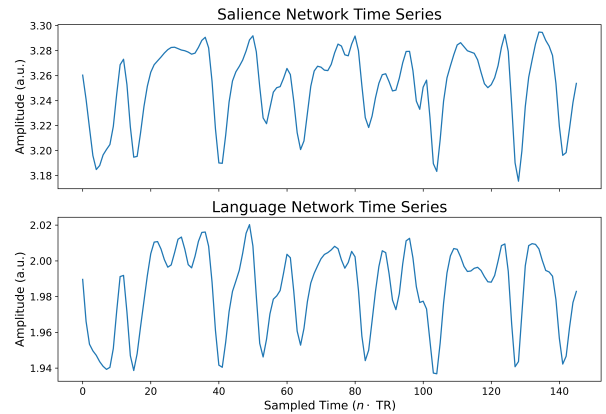


Figure 2: An example of two network time series with a high Pearson correlation coefficient for subject ID 29228, a 9-year-old male healthy control. The used repetition time is 2 seconds.

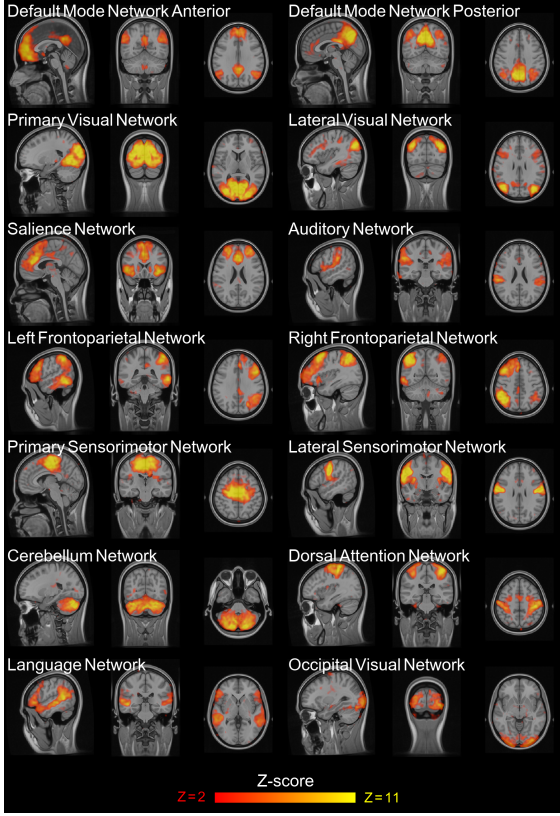


Figure 3: Visual representation of the RSNs found in the ABIDE dataset. The Z-score represents the level of correlation between a voxel and a network [Schielen et al., 2024a].

where x, y are the input time series, $\bar{x}, \bar{y}$ are the corresponding mean values and N being the length of the time series. The function provides as an output a $\mathbb{R}^{14 \times 14}$ matrix with all the correlation coefficients. To create a feature vector, the upper triangle of the matrix is transformed into a feature array of length 91, where each entry can be traced backwards to find the indices of the correlated time series. The choice between using the upper or lower triangle is arbitrary, as the output matrix is symmetric and both triangles contain identical values.

Additionally, SCC is defined as the PCC computed between the ranked variables of the time series values, making it a non-linear method. This procedure allows it to capture monotonic trends [Spearman, 1904]. SCC-based features were extracted following the same methodology as for PCC, serving as a benchmark to identify the best-performing classifier.

The final considered feature type is the DTW distance between 2 time series. The algorithm to find the distance $DTW(x, y)$ between two time series x, y is

performed in the following manner:

$$\begin{aligned}
 D(i, j) &= d(x_i, y_j) + \\
 &\quad \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\} \\
 D(0, 0) &= 0, D(i, 0) = \infty, D(0, j) = \infty \\
 d(x_i, y_j) &= (x_i - y_j)^2 \\
 DTW(x, y) &= D(N, M),
 \end{aligned} \tag{2}$$

where the length of each time series is N and M correspondingly. An l_2 normalization of the DTW feature vectors has been applied, to ensure robust comparability across subjects and mitigate the effects of heterogeneous scanning protocols. Similarly to PCC/SCC, DTW is a non-directional method and results in a symmetric output matrix. The 91-feature array is constructed from the upper triangle, excluding the diagonal. Correspondingly, the original pairs of correlated time series can be identified by their indices in the feature array.

3.3 Feature Selection

The applied feature-selection strategy is a forward sequential selection. This feature selection algorithm provides the most informative subset of features for the model. The process is initialized with the single most dominant feature and iteratively adds the next most contributive feature until 10 features have been selected. At each iteration, given a subset of selected features, the next additional feature is chosen so that the new set achieves the best performance [Kamalov et al., 2024]. This approach was chosen to find a set of strongly correlated features that can support a clinical interpretation for ASD classification.

3.4 Classification

The features of each participant were labeled with ASD or HC to allow supervised learning. Several machine learning algorithms were applied to identify the most effective classifier for the feature space, including Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbor (KNN) and Neural Network (NN). The classifiers were trained on the features data and then assessed.

To achieve a more reliable and generalized model, a 5-fold cross validation was applied. This method divides the data into random 5 sub-datasets, where for each training iteration one subset is used as the test set and the remaining subsets are used as the training set. Within each fold, an inner 3-fold CV guides feature addition and prevents data leakage by ensuring each subset is evaluated on unseen data. The

Table 1: Top classifier performance across different metrics and datasets, best results are highlighted in bold

Metric	All Data			NYU Sub Dataset with Feature Selection		
	PCC	SCC	DTW	PCC	SCC	DTW
Classifier	NN	SVM	SVM	NN	RF	LR
Accuracy	61%	63%	58%	54%	54%	65%
F1 Score	61%	63%	57%	49%	54%	65%
Sensitivity	56%	55%	40%	27%	48%	62%
Specificity	65%	70%	74%	79%	60%	67%
AUC	0.61	0.62	0.57	0.53	0.54	0.67

PCC - Pearson Correlation Coefficient; SCC - Spearman Correlation Coefficient; DTW - Dynamic Time Warping; AUC - Area Under the Curve.

combination of internal validation with an outer 5-fold CV of the classifiers following feature selection is known as nested cross-validation, a method that rigorously guards against overfitting [Santana et al., 2022] [Rosenblatt et al., 2024]. The evaluation metrics of each fold is calculated and averaged to derive the final performance of the model.

The classification was initially based on all the available data. An additional classification was performed on a subset of the data, containing only scans originating from the NYU Langone Health system.

4 RESULTS

In this section, the outcomes and implementation details of the applied methodology are listed and discussed. The processes of feature extraction, feature selection, and classification are examined along with the results of the BEP extension.

4.1 Feature Extraction

The Python version used in this research is 3.11.11. To extract the PCC and SCC features, the function `corr` has been used, provided by the Python library Pandas. Since both correlation methods are non directional, the output matrix is symmetric and thus only the upper triangle of the matrix is used as features, excluding the diagonal.

For DTW, the `distance` function from the

Table 2: Feature selection recurrence across 50 folds using Spearman’s correlation coefficient & dynamic time warping methods

Correlated Time Series Feature	Count
Anterior Default Mode Network & Posterior Default Mode Network	23
Saliency Network & Language Network	13
Lateral Sensorimotor Network & Right Frontoparietal Network	11
Primary Visual Network & Left Frontoparietal Network	10
Saliency Network & Primary Visual Network	9
Posterior Default Mode Network & Dorsal Attention Network	9
Left Frontoparietal Network & Language Network	9
Primary Visual Network & Dorsal Attention Network	9
Lateral Sensorimotor Network & Auditory Network	9
Primary Sensorimotor Network & Auditory Network	9

DTAIDistance Python library computes the shortest distance that aligns 2 time series, where short distance indicates high FC and long distance indicates low FC.

4.2 Feature Selection

A sequential feature selection algorithm was applied, using `SequentialFeatureSelector` from the scikit-learn Python library.

Since both aforementioned metrics yield feature vectors in which each index corresponds to the same pair of time series, the frequency of selection for each feature was recorded to identify the most consistently chosen features. A selection-frequency analysis across all CV folds and classifier types identified the most recurrent features. Table 2 shows the 10 time-series pairs that exhibited the greatest consistency, along their recurrence frequency. Each selected pair of networks correlation serves as a basis for ASD classification. The connectivity between the anterior and posterior default mode networks (DMNs) proved to be the most informative, selected in 23 out of 50 folds. This was followed by distinguished correlations involving the saliency and language networks, the sensorimotor and frontoparietal networks, and the frontoparietal and visual sub networks (see Fig. 6).

Related studies have reported that individuals with ASD often experience communication difficulties, which are associated with decreased FC between the anterior and posterior DMNs, supporting the findings of the present research (see Fig. 4 [Padmanabhan et al., 2017, Di Martino et al., 2014]. FC between the saliency and language networks, identified in this study as highly informative, has likewise been reported to show decreased connectivity in ASD, further reinforcing the results [Bernas, 2020]. Moreover, decreased connectivity between the lateral sensorimotor and right frontoparietal networks has also been documented in the literature, in agreement with the findings [Minnigulova et al., 2025].

The classification models trained on the chosen feature subset were evaluated and their performance

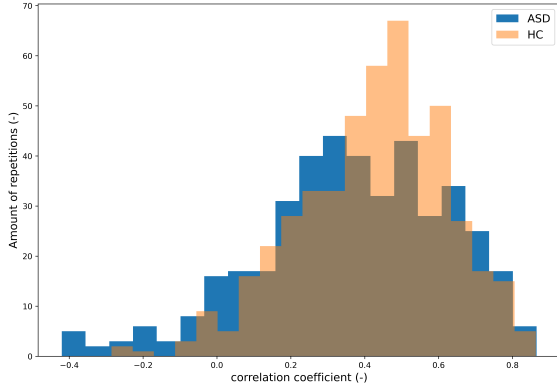


Figure 4: A comparison of the distribution of Spearman's correlation coefficient between the anterior and posterior default mode networks, among Healthy Controls (HCs) and Autistic Spectrum Disorders (ASDs) across the whole dataset.

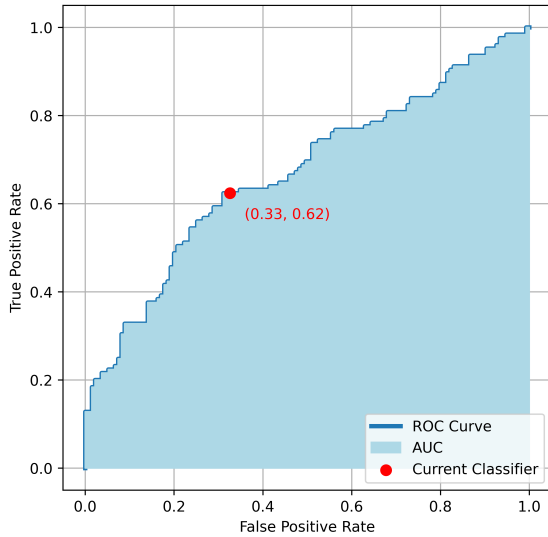


Figure 5: Receiver Operating Characteristic (ROC) curve of the dynamic time warping based classification on the NYU sub-dataset, representing the highest-performing model, with an achieved Area Under Curve (AUC) of 0.666.

benchmarked against models trained without any feature selection.

4.3 Classification

The classification was performed using the Python library scikit-learn. Across all metrics, the applied classifiers shared the same configurations. LR was set with a maximum of 1000 iterations, while the SVM used a linear kernel. The RF was assembled of 100 decision trees, KNN included 5 neighbors and the neural networks consisted of a single hidden layer with 100 perceptrons, a maximum of 1000 iterations

and a learning rate of 0.005. Table 1 represents the assessment of the examined classifiers. Initial classification was performed using all examined metrics. Among them, SCC achieved the highest classification accuracy of 63% (AUC = 0.62) using an SVM classifier, followed by PCC with an accuracy of 61% (AUC = 0.61). DTW features achieved the best result with an SVM classifier reaching an accuracy of 58% (AUC = 0.57). Based on these results, it can be concluded that ASD classification using the linear monotonic relationships (SCC) of all examined RSNs across the entire dataset outperformed the other observed metrics.

After evaluating the classification performance using all features, the performance based on the selected features was subsequently examined. The highest performance was achieved using PCC features with a Logistic Regression classifier, reaching an accuracy of 59% (AUC = 0.58). SCC and DTW-based classifications yielded lower accuracies, with maximum values of 58% and 57%, respectively. Applying feature selection to the entire dataset resulted in decreased classification performance across all metrics. This decline in performance suggests there is no subset of features that sufficiently describes the heterogeneity inherent to ASD, as also observed by [Masi et al., 2017].

Finally, classification performance based on the selected features was evaluated using the NYU sub-dataset, allowing analysis on a more homogeneous dataset. This evaluation revealed that DTW-based features yielded the highest performance, achieving an accuracy of 65% (AUC = 0.67) with a LR classifier (see Fig. 5). Following that, SCC-based features reached an accuracy of 54% (AUC = 0.54) using an RF classifier, while PCC-based features resulted in slightly lower performance, with an accuracy of 54% (AUC = 0.53) using a NN classifier. This result suggests that non-linear connectivity, when using homogeneous fMRI scans, can outperform linear connectivity in classification.

5 CONCLUSIONS

Autism spectrum disorder diagnosis remains a challenging task and an objective diagnosis is desired. This study aimed to address this challenge by evaluating multiple brain connectivity metrics - PCC, SCC, and DTW - for classifying ASD using resting-state fMRI scans. While modest classification performance was achieved overall, DTW features showed better results than the rest of the examined metrics. The best performance (65% accuracy) was achieved using DTW-based features with a Logistic Regression

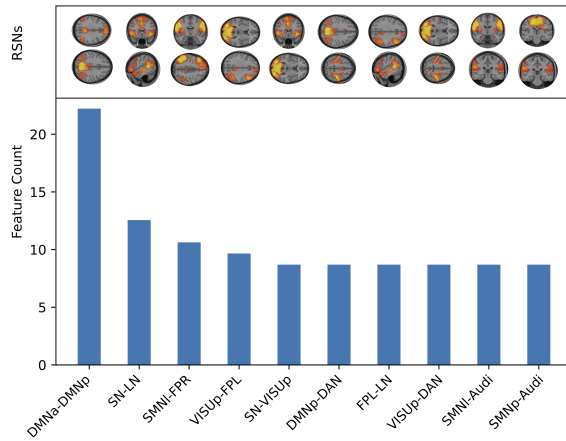


Figure 6: A bar chart that illustrates the count of selected features and the corresponding correlated Resting State Networks (RSNs), ranked by feature count in the feature selection procedure.

classifier on the large homogeneous NYU sub dataset (including 260 subjects).

Feature selection did not consistently improve performance across the full dataset, likely due to noise and site-related variability in the data. However, selected features performed better only when using DTW features with the homogeneous scanning protocol NYU sub dataset, which highlights the effectiveness of non-linear correlation methods under consistent scanning protocols. Additionally, the brain networks identified as informative align with findings from clinically related literature on autism diagnosis.

These findings demonstrate that non-linear correlation measures such as DTW can offer a meaningful advantage over traditional linear methods. They also reinforce the value of nested cross-validation and homogeneous scanning protocol data sources in developing generalizable classifiers.

Nonetheless, this study has several limitations. The method shows reduced performance when applied to heterogeneous datasets with varying scanning protocols, which may affect its generalizability. In addition, the calculation of DTW distances is approximated due to computational complexity, which may lead to reduced alignment precision. Moreover, the highest diagnostic accuracy achieved falls below the threshold typically required for clinical application. Future research should explore the integration of alternative brain atlases and evaluate more advanced classification frameworks such as deep neural networks or transformers to further enhance diagnostic accuracy.

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